

Abdullah Al Zahid

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Talent Acquisition

Rail & Transit

WSP Canada, Toronto, ON

Re: Senior Passenger Modelling Specialist (Job ID 94275)

Dear Hiring Manager,

I would like to apply for the Senior Passenger Modelling Specialist position with the Rail & Transit team, based in Toronto. Simulation has been the common thread through my work so far, and passenger modelling is where I would like to take it.

The posting asks for experience in agent-based simulation within transportation. My master's research at the University of Alberta used MATSim, an agent-based model, alongside macroscopic LWR models to study how road networks perform during emergency evacuations: how many people move, which paths they take, where queues form and whether the system holds up under stress. Those are close to the questions behind passenger flow modelling, though at network rather than station scale. Since then I have run traffic simulations of Ontario Line construction staging at Mott MacDonald, and OpenTrack simulations of GO rail services at Metrolinx, building and testing scenarios against infrastructure constraints before anyone committed to them.

The posting also asks for strong Python and Excel for analysis and automation. At Metrolinx I set up a command line workflow that pre-simulated slow orders in batches, cutting turnaround on scenario requests from days to hours, and built the Crew Variance Tool, a Python application that read large JSON operations files and summarised speed profiles and run-time variance across subdivisions. At the City of Toronto I built the Transportation Data Dashboard, bringing several years of data into one validated source, and I write the technical reports and presentations that turn analysis into something decision-makers can act on.

I should be straightforward about the gaps. I have not used LEGION, Viswalk, MassMotion or CAST, modelled a transit station, or developed a Crowd Control Plan, and I have not used VBA or R. The posting asks for eight or more years of experience; I have closer to four years of professional work plus my graduate research. I am applying because the modelling foundation is genuinely mine, and if the team has an intermediate passenger modelling role, I would welcome being considered for it.

Thank you for considering my application.

Sincerely,

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